

Vasu Agarwal

+91-8126537874 | avasu769@gmail.com | [linkedin.com/in/vasuagarwal](https://www.linkedin.com/in/vasuagarwal) | github.com/VasuAgarwal22

Education

B.Tech in Computer Science and Engineering (AKTU)

ABES Engineering College

2024 – 2028

Ghaziabad, Uttar Pradesh

- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Web Technologies, Software Engineering

Class 12th (CBSE)

K.L International School

[2024]

[92.6]

Class 10th (CBSE)

K.L International School

[2022]

[91.6]

Technical Skills

- **Programming Languages:** Java, C, C++, JavaScript, SQL, Python (learning)
- **Frameworks and Libraries:** Spring Boot, Spring MVC, Hibernate ORM, Maven, React (learning), Thymeleaf, Tailwind CSS
- **Databases:** MySQL, PostgreSQL
- **Core Competencies:** Data Structures and Algorithms, Object-Oriented Programming, RESTful API Development, MVC Architecture, ORM, Time and Space Complexity Analysis
- **Developer Tools:** Git, GitHub, LeetCode, HackerRank

Experience

Independent Projects and Academic Development

Java Backend Developer

2024 – Present

Ghaziabad, Uttar Pradesh

- Built and deployed 4+ backend projects using Java, Spring Boot, Hibernate ORM, and relational databases; version-controlled all code on GitHub.

Projects

Smart Bank Application | Java, Hibernate ORM, PostgreSQL, Maven | [GitHub](#)

2024 – Present

- Developed a banking backend application using Java, Hibernate ORM, and PostgreSQL to manage customers, accounts, transactions, and ATM card operations.
- Implemented JPA/Hibernate entity relationships including One-to-Many and One-to-One mappings for Customer, Account, Transaction, and ATM Card modules.
- Built transaction-safe banking operations such as deposit, withdrawal, and fund transfer using Hibernate transaction management with commit and rollback mechanisms.
- Designed a layered architecture separating UI, business logic, and persistence layers, ensuring maintainable and scalable backend development.

File Hider CLI Tool | Java, File I/O, System Commands | [GitHub](#)

2024

- Built a Java CLI tool that toggles OS-level system attributes to hide and unhide files; reduced manual file management steps by 60%.
- Added exception handling and cross-platform command support covering 2+ operating system environments.

Sorting Algorithm Visualizer | Java | [GitHub](#)

2024

- Built a Java console visualizer for 3 sorting algorithms (Bubble, Selection, Insertion) tracking comparison counts, swap counts, and execution time in real time.
- Benchmarked all 3 algorithms across 5+ input sizes; demonstrated $O(n^2)$ vs optimized complexity trade-offs.

Competitive Programming and Achievements

LeetCode

457 Problems Solved

2026 – Present

- Solved 457 problems (Easy, Medium, Hard) across 7+ topic areas: arrays, strings, linked lists, trees, graphs, dynamic programming, and sliding window.
- Rewrote 50+ brute-force solutions into optimized approaches, cutting average time complexity from $O(n^2)$ to $O(n \log n)$ or better.

HackerRank | Problem Solving Certified

2024

- Earned HackerRank Problem Solving certification — top 10% of certified candidates in algorithmic thinking and data structures.

Additional Information

- **Languages:** English, Hindi
- **Interests:** Backend engineering, full-stack development, open-source contribution, competitive programming, SWE internships